



VOKE AI

PLATFORM ENGINEERING GROUP

ARCHITECTURE WHITEPAPER

PRODUCTION READY

— ENTERPRISE TECHNICAL SPECIFICATION

System Architecture & Complete API Ecosystem

Comprehensive engineering blueprint cataloging every external and internal API, real-time pipeline, multimodal vision engine, polyglot execution sandbox, and automated job intelligence service powering the Voke platform.

24+

Integrated APIs

LLM, Multimodal Vision, STT/TTS, Polyglot Sandbox, 7 Job Feeds & Payment Webhooks.

8

Core Pipelines

Live voice WebSocket, video body language, automated code verification & career plans.

<250ms

Streaming SLA

Sub-second inference responses with multi-key failover and client-side sandboxing.

DOCUMENT SCOPE

Full Platform API Architecture & Data Flows

AUDIENCE

Executive Leadership & Core Engineering

VERSION

v2.4.0 (Enterprise Specification)

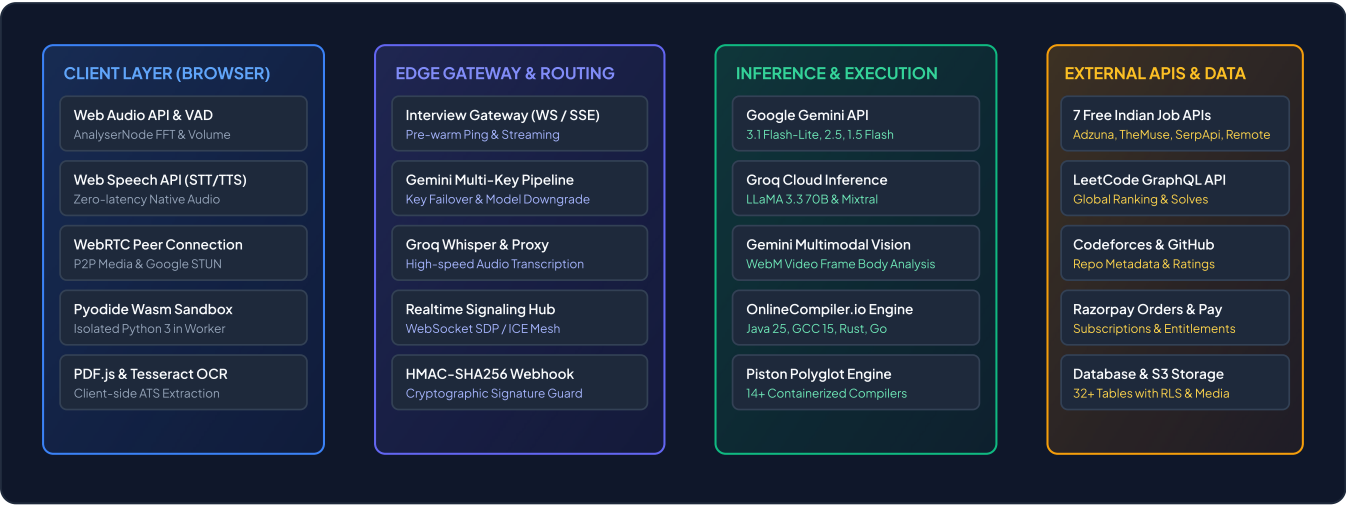
CLASSIFICATION

CONFIDENTIAL • PROPRIETARY

MACRO SYSTEM TOPOGRAPHY

Multi-Tier API Ecosystem & Micro-Services

Voke operates a resilient, distributed, low-latency microservice architecture designed for synchronous interactive interviewing, multimodal video posture grading, real-time code evaluation, and automated career intelligence.



Streaming Gateway

WebSocket / SSE Engine

- Multiplexes voice audio and token streams via WebSockets with <100ms first-byte latency.
- Pre-warm ping protocols reduce initial cold-start model latency to zero.

Multi-Key Failover

Zero-Downtime Key Pool

- Dynamic key rotation across primary, professional, and auxiliary quotas on 429/403.
- Cascading model downgrades ensure interview continuity under outages.

Hybrid Sandboxing

Wasm + Container Hybrid

- Python 3 runs locally in browser Web Workers via Pyodide Wasm—zero cloud cost.
- Compiled languages (C++, Java 25, Rust) execute inside isolated cloud runtimes.

FUNCTIONAL DOMAIN	PRIMARY TECHNOLOGY / PROTOCOL	SLA / LATENCY TARGET	RESILIENCE STRATEGY	SECURITY & ISOLATION
Live Voice Interview	WebSocket + Gemini SSE Stream	< 250 ms	Multi-Key Rotation + Groq Fallback	JWT Authentication + Turn Constraints
Video & Posture AI	Gemini 2.5 Flash Multimodal Vision	1.8 s - 2.5 s	Inline Base64 + S3 Storage Fallback	Private S3 Bucket + User Path RLS
Code Evaluation	Client Wasm + OnlineCompiler.io	0 ms (Local) / 800 ms (Cloud)	Piston Container Engine Fallback	Thread Sandboxing & 3000ms Timeout
Job Scouting	7 Free Public Aggregator APIs	Async Batch Ingestion	Deduplication & Source Failover	Sanitized Queries & Domain Locks
Monetization	Razorpay API + Webhooks	Instant State Sync	Idempotent DB Upsert on Event	HMAC-SHA256 Web Crypto Signature

CONVERSATIONAL INTELLIGENCE & EVALUATION

Multi-Model LLM Inference Pipeline

Voke utilizes a tiered, multi-provider generative AI strategy combining Google Gemini's advanced reasoning and structured JSON output with Groq's high-speed inference for conversational sub-second responsiveness.

Google Gemini Multi-Key Dynamic Failover Pipeline

High Availability • Zero Rate-Limit Dropouts

STEP 01
Client Request
WebSocket
Payload

STEP 02
Context Build
Mandates +
History

STEP 03
Key Execution
Gemini 3.1 Lite

STEP 04: 429/403
Key Rotation
Rotate Key (0 → N)

STEP 05: ON 5XX
Model Cascade
2.5 Flash → 1.5

STEP 06
Chunk Stream
SSE / WebSocket

PROVIDER & API ENDPOINT	MODELS DEPLOYED	INTEGRATION MODE	ARCHITECTURAL ROLE & RESPONSIBILITIES	OUTPUT PROTOCOL
Google Generative Language API /v1beta/models/{model}:generateContent /v1beta/models/{model}:streamGenerateContent	gemini-3.1-flash-lite gemini-2.5-flash gemini-1.5-flash	REST & SSE Server-Sent Events	Conversational Technical Interviewer: Generates live adaptive questions, verifies candidate algorithmic approaches, conducts FAANG-tier Round 3 coding dialogues, synthesizes 6Q personality scorecards, and formulates 3-month career roadmaps.	Streaming Text Chunks & Structured JSON Schema (responseMimeType: application/json)
Groq Cloud Inference API /openai/v1/chat/completions	llama-3.3-70b-versatile mixtral-8x7b-32768	REST High-Speed Inference	Sub-Second Conversational Agent & ATS Parser: Performs ultra-fast answer evaluations during mock sessions, executes high-accuracy ATS resume parsing into typed JSON schemas, and computes vector similarity for candidate job recommendations.	Direct JSON Object (type: json_object) & Plain Text
Lovable AI Gateway API /v1/chat/completions	gemini-2.5-flash gpt-4o-mini-proxy	REST Cloud Proxy	Secondary Inference Gateway: Serves as an auxiliary fallback route for analyzing community interview trends, generating career guidance, and evaluating student placement mock rounds if primary upstream routes saturate.	JSON Payload Response

Dynamic Key Rotation & Cascade Engine

99.99% Availability

- Multi-Key Pool: Dynamically rotates across primary, pro, and dynamic keys.
- Instant 429/403 Recovery: Key indices increment seamlessly without dropping candidate sessions.
- Graceful Model Downgrade: If 5xx errors occur, requests step down: 3.1 Lite → 2.5 Flash → 1.5 Flash.

6Q Personality Evaluation Framework

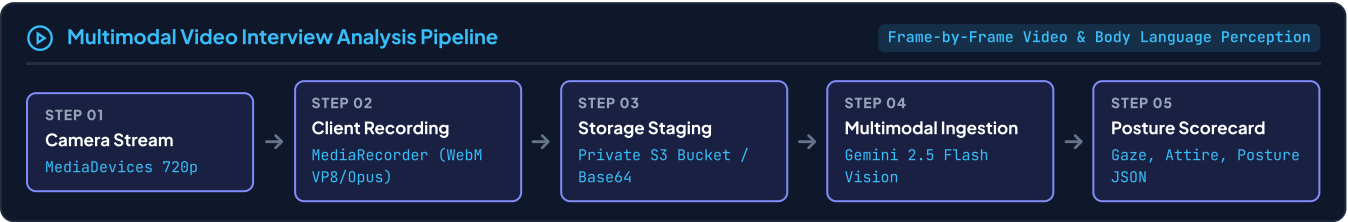
JSON Schema Locked

- 6Q Metric Dimensions: Scored across IQ, EQ, CQ, AQ, SQ, and TQ (0-100 scale).
- Strict Output Lock: Guaranteed syntax-safe JSON schemas eliminate radar chart rendering bugs.
- Anti-Repetition Engine: Prevents asking repeat or rephrased questions during interviews.

VISUAL BODY LANGUAGE & GAZE PERCEPTION

Multimodal Video Analysis Pipeline

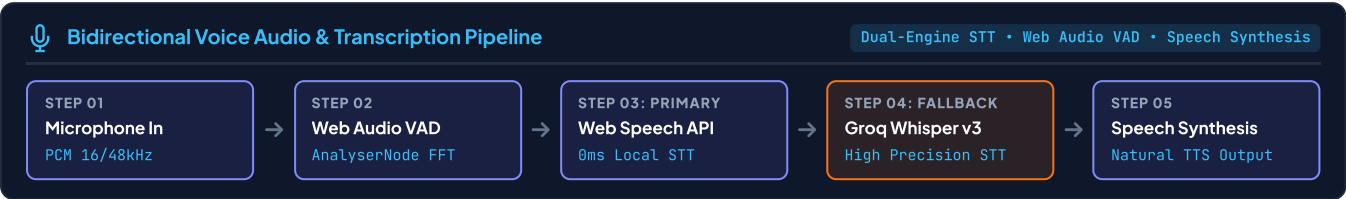
Voke utilizes Google Gemini's advanced multimodal video vision capabilities to inspect candidate physical posture, eye-contact stability, hand gesturing, facial affect, and professional attire directly from recorded video streams.



REAL-TIME VOICE PIPELINE

Bidirectional Speech & Voice Activity Systems

Voke couples native browser speech APIs with cloud-hosted Groq Whisper models to provide resilient, low-latency, domain-biased voice recognition and conversational speech delivery.



VOICE SUBSYSTEM	API / SPECIFICATION	INPUT / OUTPUT PROTOCOL	OPERATIONAL CHARACTERISTICS & TECHNICAL CAPABILITIES	LATENCY / SLA
Primary STT	Web Speech API (SpeechRecognition) window.webkitSpeechRecognition	Continuous PCM audio stream → String text	Runs 100% locally on user machine with zero network bandwidth consumption. Emits interim word hypotheses and triggers speech finalization upon conversational pause.	< 50 ms (Zero Network Hop)
Fallback STT	Groq Cloud Whisper API whisper-large-v3-turbo	Multipart audio/webm blob → JSON transcript	High-fidelity transcription for complex tech accents. Runs with temperature 0, English lock, and domain prompting ("Technical software engineering interview speech strictly in English").	200 - 350 ms (Cloud Inference)
Voice Activity (VAD)	Web Audio API (AudioContext) AnalyserNode & createMediaStreamSource	MediaStream → Float32 frequency data	Computes root mean square (RMS) volume metrics, dynamically filters ambient room noise, drives audio visualizer waveforms, and triggers barge-in silence detection.	Real-Time (Audio Thread)
Interviewer TTS	Web Speech Synthesis API SpeechSynthesisUtterance	Clean text string → Hardware speaker audio	Delivers natural spoken dialogue. Features automated resume heartbeat to bypass Chromium synthesis pause bugs, plus instant barge-in cancel when candidate starts speaking.	Instantaneous (Local Playback)

Candidate Coding Focus Protection: The Silent Mode Protocol

Round 3 Algorithmic Assessment

Engineered Interruption Prevention Architecture

- Automated Audio Muting:** When candidate transitions to Round 3 (Algorithmic Live Coding) or types in the code editor, the client orchestrator enforces *Silent Mode*. Microphone recording stops, speech synthesis cancels, and AudioContext suspends.
- Approach Verification Gate:** Once the candidate verbally explains their algorithmic approach (>50% relevance), the interviewer responds [APPROACH_VERIFIED], unlocks the editor, and remains completely silent until tests run.

ZERO-TRUST SAFE CODE EXECUTION

Polyglot Multi-Language Sandbox Architecture

Voke implements a hybrid sandboxing architecture: Python and TypeScript execute locally inside WebAssembly and Web Worker isolation with zero server cost, while compiled enterprise languages (C++, Java, Go, Rust) execute via secure containerized APIs.



Automated Test Harness Validation

Zero Regressions

- Structured Test Feeds:** Wraps candidate code with test harnesses feeding complex vectors (trees, graphs, matrices) via stdin.
- Normalization:** Trims trailing whitespaces and normalizes numeric/string types to guarantee test comparisons evaluate accurately.

Algorithmic Complexity Evaluation

Round 3 Post-Run

- Big-O Interrogation:** Once tests pass, the AI questions candidate on Big-O Time Complexity and Space Complexity tradeoffs.
- Telemetry Logging:** Records execution milliseconds and memory usage into candidate performance reports.

REAL-TIME INDIAN & GLOBAL TECH EMPLOYMENT FEEDS

7 Integrated Public Job Discovery Feeds

Voke continuously aggregates everyday live technology postings across 7 public APIs and scrapers specifically focused on the Indian Job Market and remote engineering opportunities.

JOB SOURCE & PROVIDER	ENDPOINT & QUERY SIGNATURE	GEOGRAPHIC FOCUS	AUTH REQUIREMENT	INGESTION & PARSING PROTOCOL
Adzuna Jobs API India Targeted	<code>api.adzuna.com/v1/api/jobs/in/search</code>	India Region (in)	Optional App ID/Key (10,000 req/mo)	Extracts structured Indian tech postings with normalized salary bands (₹ INR), company name, and location filters.
Google Jobs via SerpApi India Targeted	<code>serpapi.com/search.json?engine=google_jobs&gl=in</code>	India Regional Search (gl=in)	SerpApi Key (Free Tier 100/mo)	Extracts verified employer listings indexed by Google Jobs across Indian metropolitan tech hubs (Bengaluru, NCR, Hyderabad, Pune).
The Muse Public API 100% Free Public	<code>themuse.com/api/public/jobs?location=India&category=IT</code>	India & Global Remote	✗ No Auth Required	Paginates Software Engineering, Data Science, and IT listings with rich corporate culture context and benefits data.
RemoteOK Public API 100% Free Public	<code>remoteok.com/api?location=india</code> <code>remoteok.com/api?tags=cybersecurity</code>	Global & India Remote	✗ No Auth Required	Extracts developer, DevOps, and Cybersecurity roles filtered by technology tags, salary markers, and direct apply links.
Remotive API 100% Free Public	<code>remotive.com/api/remote-jobs?search=india</code>	Worldwide Remote & APAC	✗ No Auth Required	Pulls active remote engineering openings, technology requirements, and candidate location constraints.
Jobicy API 100% Free Public	<code>jobicy.com/api/v2/remote-jobs?geo=apac</code>	APAC & India Region	✗ No Auth Required	Fetches remote technology postings specifically open to candidates located in the Asia-Pacific region.
Findwork.dev API 100% Free Public	<code>findwork.dev/api/jobs/?location=india</code>	India Developer Market	✗ No Auth Required	Extracts software engineering openings tagged by programming language and experience level.

Targeted Geographic & Metro Scouting

India Active Filter

- Dynamic Monitored Locations:** Edge ingestion queries database table of active locations (Bengaluru, Delhi NCR, Mumbai, Hyderabad, Pune, Chennai).
- Regional Localization:** Formats location parameters (loc, India) across all external providers to capture domestic opportunities.

Specialized Cybersecurity & IT Domains

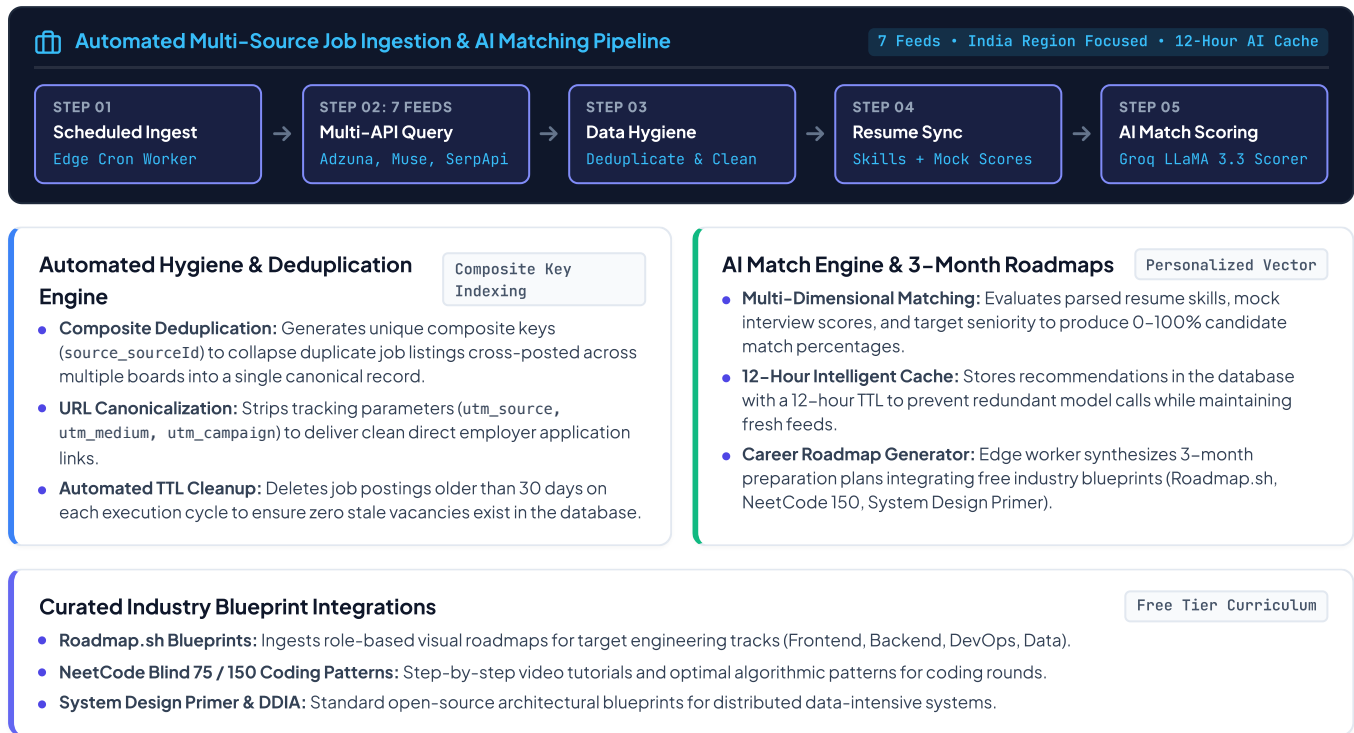
Domain Ingestion

- Specialized Queries:** Automatically queries designated domain verticals: SOC Analyst, Cybersecurity, Desktop Support, and Information Security.
- Tag Filtering:** Filters in-memory feeds for domain keywords to populate specialized cybersecurity tracks.

TALENT MATCHING & PREPARATION ROADMAPS

Data Hygiene, AI Matching & Career Engine

Voke's background workers scrub and deduplicate multi-source feeds, match vacancies against candidate resume skills, and generate personalized 3-month career preparation roadmaps.



ALGORITHMIC READINESS & PROJECT VERIFICATION

External Profile Verification & Context Pipeline

Voke integrates directly with LeetCode, Codeforces, and GitHub public APIs to verify candidate coding skills, contest ratings, solved problem metrics, and extract actual repository code context for personalized project deep-dive interviews.

🔍

Candidate Profile Verification & Project Context Pipeline

GraphQL & REST Ingestion • Verified Skill Badges

STEP 01
Profile Handles
LeetCode / CF / GitHub

→

STEP 02: LEETCODE
GraphQL Profile
Solves by Difficulty + Rank

→

STEP 03: CF
REST User Info
Rating & Rank Check

→

STEP 04: GITHUB
README Decode
Base64 Project Decode

→

STEP 05
Interview Context
Inject Project Deep-Dives

PLATFORM & PROTOCOL	ENDPOINT & PAYLOAD STRUCTURE	DATA POINTS EXTRACTED	PLATFORM UTILIZATION & PIPELINE IMPACT
LeetCode Official API GraphQL POST	POST https://leetcode.com/graphql Query: getUserProfile(\$username: String!)	• submitStatsGlobal: Easy, Medium, Hard solves. • userContestRanking: Contest rating, global rank, top percentile.	Algorithmic Verification: Displays verified problem badges; sets starting question difficulty in technical mock interviews.
Codeforces Official API REST GET	GET /api/user.info?handles={handle} GET /api/user.status?handle={handle}	• Current & max competitive programming rating. • Official rank title. • Recent accepted submissions (verdict: "OK").	Solution Verification: Validates candidate solutions to contest problems and awards verified competitive programming badges.
GitHub REST API REST GET	GET /users/{username}/repos GET /repos/{user}/{repo}/readme	• Top repositories sorted by updated date. • Stars, language distribution, forks. • Project README markdown decoded from base64.	Project Deep-Dive Rounds: Parses repository READMEs to instruct the AI to challenge candidate on their actual open-source projects.

Project Deep-Dive Synthesis Engine

Context Injection

- **README Decoding:** Decodes base64 README content, strips markdown formatting, and extracts core technology stacks and architectures.
- **Targeted Questions:** Challenges candidate on architectural decisions: *"In [Project], why did you choose PostgreSQL over MongoDB, and how did you handle caching?"*

Anti-Fraud Solution Verification

Real-Time Validation

- **Live Profile Polling:** When candidate marks an external problem as solved, the gateway queries the external platform API.
- **Timestamp Check:** Verifies an accepted submission (verdict === "OK") exists under the candidate's verified handle.

Voke Platform Architecture • Section 08

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ENTERPRISE BILLING & WEBHOOK INGESTION

Cryptographic Payments & Entitlement Architecture

Voke integrates Razorpay's end-to-end payment ecosystem with server-side price calculation, client modal checkout, and Web Crypto API HMAC-SHA256 cryptographic signature validation for idempotent subscription provisioning.

Cryptographic Payment Creation, Checkout & Webhook Ingestion Pipeline

HMAC-SHA256 • Server-Authoritative Pricing • Idempotent

STEP 01
Plan Selection
User JWT +
Coupon Code

STEP 02
Order Creation
Razorpay Orders
(Paise)

STEP 03
Client Checkout
Razorpay Modal

STEP 04: 2
ROUTES
Verification / Ho...
order.paid /
captured

STEP 05
HMAC-SHA256
Subtle Crypto
Check

STEP 06
Entitlement Grant
DB Subscription
Upsert

PAYMENT SUBSYSTEM	API ENDPOINT & PROTOCOL	AUTHENTICATION / SECURITY	PAYLOAD & PARAMETER SPECIFICATIONS	DATABASE ACTION
Order Creation Gateway Server-to-Server	POST https://api.razorpay.com/v1/orders	HTTP Basic Auth (btoa(KEY_ID:KEY_SECRET))	Locks plan cost on backend (e.g. ₹399 = 39900 paise). Computes dynamic discount coupons (e.g. 30% discount → ₹279 = 27900 paise). Encodes user ID and receipt in notes.	Generates order entity bound to user ID; blocks client tampering.
Client Modal Checkout Client SDK	checkout.razorpay.com/v1/checkout.js	Publishable Razorpay Key ID	Mounts Razorpay Standard Modal. Accepts order_id, pre-fills candidate email and contact, presents UPI, NetBanking, and Card instruments.	Returns payment signature payload upon success.
Client Payment Verification Verification Edge	POST /functions/v1/verify-razorpay-payment	Bearer User JWT + Server Secret	Validates signature: HMAC_SHA256(order_id + " " + payment_id, SECRET) using Web Crypto API. Constant-time byte matching eliminates timing attacks.	Updates candidate subscription to is_premium = true and grants premium tier.
Async Webhook Listener Server-to-Server	POST /functions/v1/razorpay-webhook	x-razorpay-signature Header Check	Ingests order.paid and payment.captured events. Validates raw body signature before JSON parsing to protect against payload tampering.	Executes idempotent upsert with user ID and payment ID to prevent duplicate records on retry.

Timing-Resistant Signature Check

Web Crypto API

- Native Cryptography:** Uses crypto.subtle.importKey and crypto.subtle.sign("HMAC", ...).
- Constant-Time Comparison:** Employs bitwise XOR across hash length, eliminating timing vulnerabilities.

Replay Attack & Idempotency Protection

Database Constraint

- Unique Order Binding:** Each order receipt encodes timestamp and user ID (rcpt_{userId}_{timestamp}).
- Duplicate Webhook Defense:** Database executes an atomic upsert on conflict with user_id.

REAL-TIME MESH, DOCUMENT PARSING & DATA LAYER

WebRTC P2P Mesh & Core Infrastructure APIs

Voke operates browser-to-browser WebRTC media streams coordinated through WebSocket signaling channels for peer mock interviews, alongside client-side spatial PDF parsing, geolocation, and a hardened relational database.

SUBSYSTEM	PROTOCOL / API ENDPOINT	UNDERLYING ENGINE / SERVICE	ARCHITECTURAL FUNCTION & FLOW	RESILIENCE / FALLBACK STRATEGY
WebRTC P2P Video Mesh Real-Time P2P	RTCPeerConnection Google STUN (stun.l.google.com:19302)	Browser WebRTC Engine	Connects candidates in 1-on-1 peer mock interviews with low-latency bidirectional audio/video streams. Google STUN handles NAT traversal and ICE candidate discovery.	Multiple fallback STUN endpoints (stun1 through stun4) ensure connection across strict firewalls.
Realtime Signaling Broker WebSocket Plane	Realtime Channels API channel("room:{sessionId}")	Supabase Realtime (WebSockets)	Broadcasts WebRTC Session Description Protocol (SDP) offer, answer, and ice-candidate payloads, alongside collaborative text chat.	Automatic WebSocket reconnection loop with heartbeat pings and presence detection.
Spatial PDF Resume Parser Client WebAssembly	Dynamic Worker Import pdfjs-dist (pdf.worker.min.mjs)	PDF.js Engine & Tesseract.js	Validates %PDF- magic bytes, extracts text elements with Y-descending and X-ascending geometric sorting to preserve multi-column reading order, and extracts link annotations.	If extracted text is under 50 characters (scanned image resume), triggers Tesseract.js client OCR Web Worker.
Relational Database Layer PostgREST API	Database REST / GraphQL API /rest/v1/{table}	Hardened PostgreSQL 15	Manages 32+ relational tables (interview sessions, video scores, job postings, career plans) protected by Row-Level Security (RLS) policies tied to caller JWTs.	Automated continuous backups, index optimization, and connection pooling.
Encrypted Object Storage S3 Compatible	Storage API /storage/v1/object/{bucket}/{path}	Private S3 Storage Buckets	Stores video interview recordings ({user_id}/{session_id}.webm) and uploaded PDF resumes. Non-public buckets enforce folder-level RLS policies.	Expiring signed URLs for authorized video analysis access; 500MB hard limit per video.
IP Geolocation & Telemetry REST GET	GET https://ipapi.co/json/	IPAPI Geolocation Service	Automatically detects candidate city and country for job search defaulting, local currency display (₹ INR vs \$ USD), and regional analytics logging.	Caches result in local storage; gracefully falls back to default region (India) if blocked.
Campus Drive Synchronization Serverless API	GET / POST /api/college-drives	Vercel Serverless Function	Synchronizes scheduled campus placement drives, question banks, and student assessment slots across institutional dashboards.	In-memory deduplication map with persistent state merging.

Spatial Layout PDF Extraction Protocol

Human Reading Order

- Coordinate Vector Sorting:** Standard parsers scramble two-column resumes. Voke evaluates X and Y coordinates to reconstruct exact reading sequences.
- Annotation Harvesting:** Reads underlying PDF link annotations to extract clean URLs for candidate GitHub, LinkedIn, and portfolio sites.

Enterprise Row-Level Security (RLS)

Zero-Trust Isolation

- User Path Protection:** All database queries and video storage buckets evaluate `auth.uid() == user_id` at kernel level.
- Institutional Boundaries:** College placement assessment scorecards are partitioned so that college admins only access their own students.

COMPREHENSIVE OPERATIONAL WORKFLOWS

Master Pipeline Blueprints & Security Architecture

Chronological data lifecycle workflows mapping the sequential interaction of client subsystems, edge gateways, inference engines, sandboxes, and persistence layers across Voke's core user journeys.

Pipeline A: Live Conversational Technical Interview Flow

WebSocket Realtime Stream

Sequence: Candidate enters room → Client initiates WebSocket handshake with pre-warm ping → Candidate speaks into microphone → Web Audio API computes VAD volume envelope → Web Speech API transcribes speech in real time (fallback: Opus chunk sent to Groq Whisper v3 Turbo) → Formatted turn context with anti-repetition mandates sent to Streaming Gateway → Gateway invokes Gemini Multi-Key Pipeline → Tokens stream via SSE chunk-by-chunk to client → Browser Web Speech Synthesis delivers spoken response → On Round 3 coding question, Silent Mode automatically mutes AI speech.

Pipeline B: Video Interview Body Language & Posture Scoring

Multimodal Vision AI

Sequence: Candidate answers question on webcam → MediaRecorder captures 720p WebM stream → Video staged to private S3 storage bucket under path-based RLS → Video payload transmitted as base64 inline stream to Gemini 2.5 Flash Multimodal Vision API → Vision engine conducts frame-by-frame analysis of eye contact, head stability, hand gestures, facial expressions, and attire appropriateness → Outputs structured JSON scorecard with delivery, posture, and confidence metrics (0-100) → Persisted to database and visualized on user report card.

Pipeline C: Multi-Source Job Scraping & Resume Matching

7 Public APIs + AI Scorer

Sequence: Scheduled Edge Cron triggers → Automated worker queries 7 public job APIs (Adzuna India, SerpApi Google Jobs, The Muse, RemoteOK, Remotive, Jobicy, Findwork) → Deletes listings older than 30 days → Deduplicates via composite source IDs and strips UTM tracking → Batches clean jobs into database → When user opens recommendations, worker extracts parsed resume skills and mock interview scores → Groq LLaMA 3.3 computes vector match percentage → Results cached for 12 hours with instant 3-month career roadmap generation.

Pipeline D: Cryptographic Payment & Entitlement Provisioning

HMAC-SHA256 Webhooks

Sequence: Candidate selects Elite Pro plan → Client invokes Order Gateway with user JWT → Server computes authoritative price in INR paisa and applies coupon discount → Creates authenticated Razorpay order entity → Client mounts Razorpay checkout modal → Upon payment success, webhook listener intercepts order.paid event → Verifies raw body with crypto.subtle HMAC-SHA256 signature in constant time → Idempotently upserts subscription record to unlock premium interview tiers.

SECURITY & RELIABILITY PILLAR	ARCHITECTURAL MECHANISM	THREAT MITIGATION	PRODUCTION GUARANTEE
Quota & Rate Resilience	Dynamic Multi-Key Pool + Model Fallback Cascade	Third-party API rate limits (HTTP 429) & outages (HTTP 5xx)	Zero interrupted interviews during peak campus drives
Execution Sandboxing	Client WebAssembly (Pyodide) + Isolated Containers	Malicious code execution, fork bombs, server resource theft	Zero server compromise; 3000ms hard timeout limit
Payment Integrity	Web Crypto HMAC-SHA256 Constant-Time Verification	Payment spoofing, replay attacks, client price tampering	100% verified transactional ledger in INR paisa
Data Privacy & Access	PostgreSQL Kernel Row-Level Security (RLS)	Unauthorized cross-tenant or cross-user data exposure	Strict path isolation for video sessions and resumes